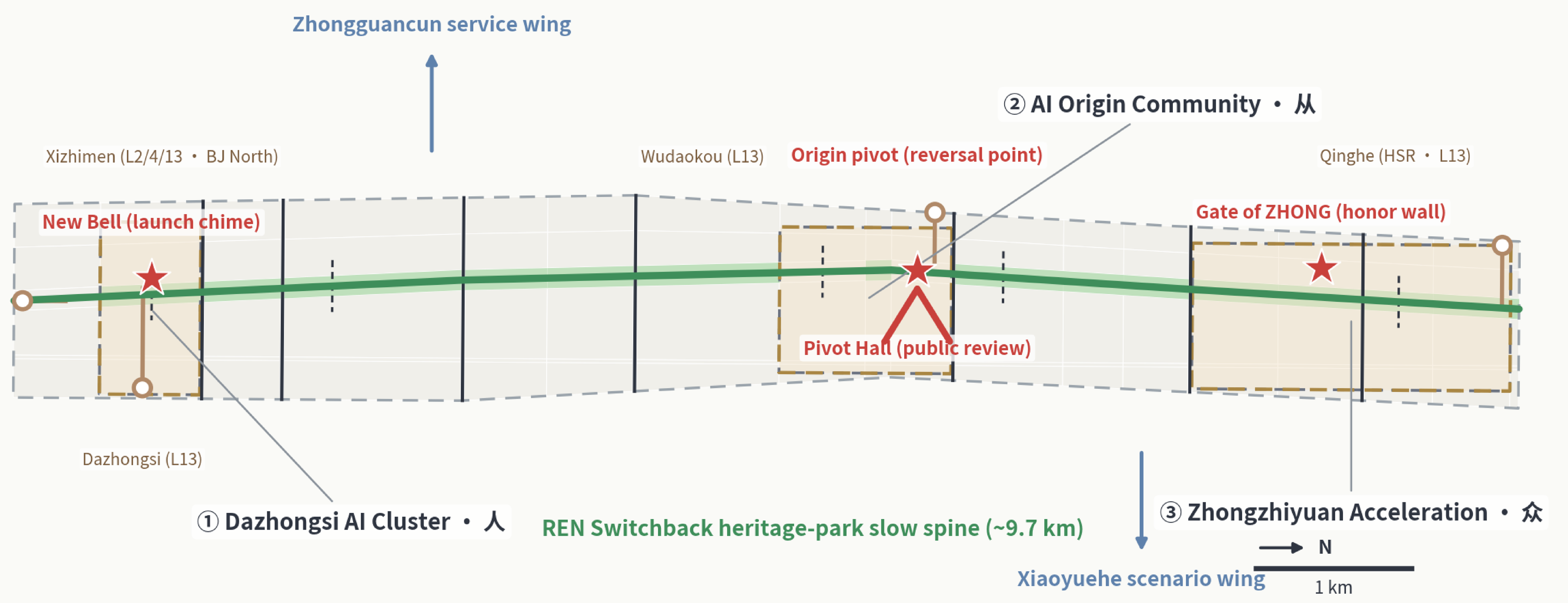
The REN Switchback — Urban Design for the Centennial Jing-Zhang AI Innovation Belt, Pivoting on 人

In 1909 at Qinglongqiao, the Jing-Zhang Railway used a switchback shaped like the character 人 to turn one unclimbable grade into two climbable ones. This proposal turns that wisdom into urban institution: one heritage-park slow-mobility spine, one origin pivot, three climbing segments (人–从–众), two coordinated wings, and twelve REN-clasp stitching crossings — every climb of intelligence passes a reversible, reviewable, roll-back-able REN pivot.

Fig.1 • Concept: The REN Switchback — an AI belt pivoting on the human

One spine • One pivot • Three grades • Two wings • Twelve stitches



In 1909 the Jing-Zhang Railway conquered an impossible grade at Qinglongqiao with a reverse switchback shaped like 人 — the character for "human". One steep line became two climbable ones; the pivot is the room engineering leaves for people. This proposal translates that wisdom into the AI belt: every climb of intelligence passes a human pivot — reversible, reviewable, revocable.

Naming system: 人 REN → 从 CONG → 众 ZHONG (one pivots • pairs collaborate • crowds accelerate)

Boundary status: provisional inference • NOT an official redline • areas recomputed in EPSG:4548 • full recomputation once official data is released
Data: geometry/*.geojson and metrics.json in this package (officially announced areas labelled separately)

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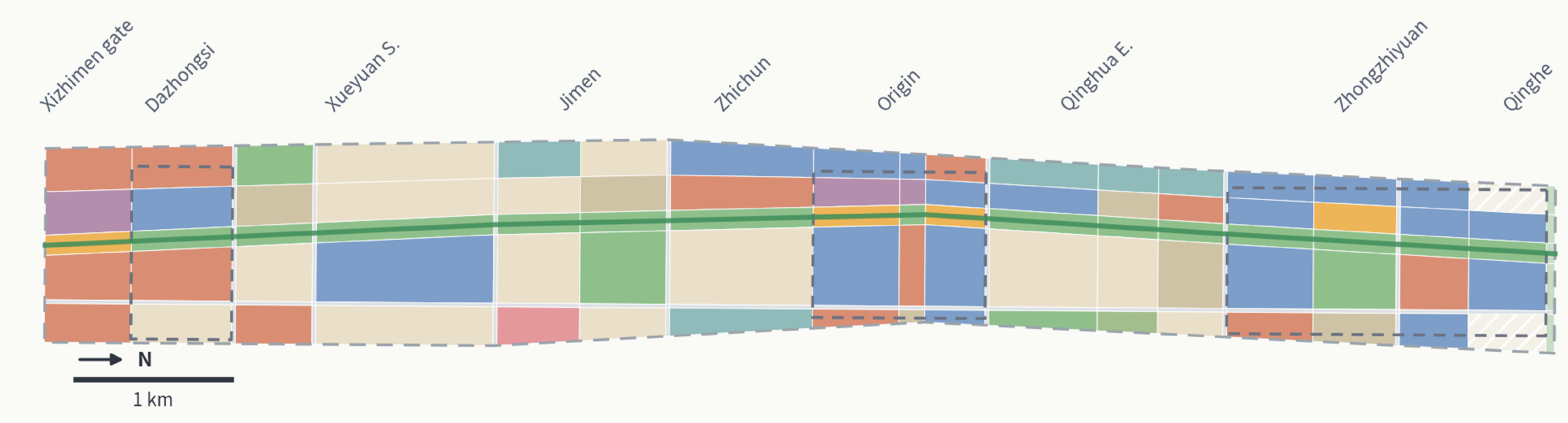
Fig.2 • Spatial structure and land use: a mixed innovation belt on a green spine

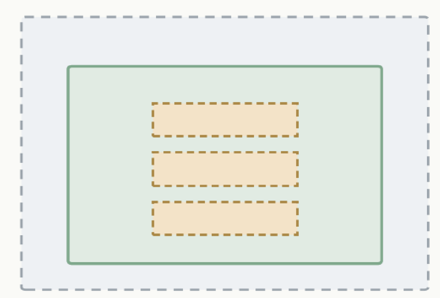
150 seamless land-use cells • six parallel bands • statutory controls pending



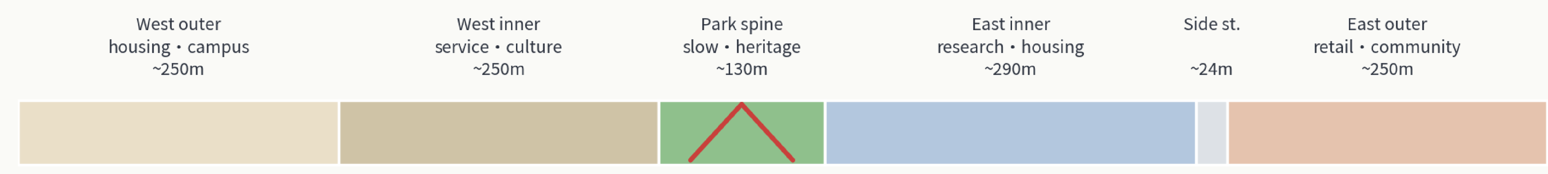
Land use (areas recomputed under prov

0802 Research	266.1 ha
05 Commercial	168.7 ha
0701 Residential	280.0 ha
0702 Community svc.	61.5 ha
0803 Culture	27.9 ha
0804 Education	51.1 ha
0805 Sports	5.2 ha
0806 Health	12.3 ha
1401 Park	169.6 ha
1402 Buffer green	5.8 ha
1403 Plaza	28.8 ha
1207 Street land	43.2 ha
16 Reserved	21.0 ha

Three nested scopes



- Research area 43.6 km²
- Overall design area 11.4 km² (this map)
- Key areas 368.4 ha ×3



Typical cross-section (west to east, six bands, indicative average widths)

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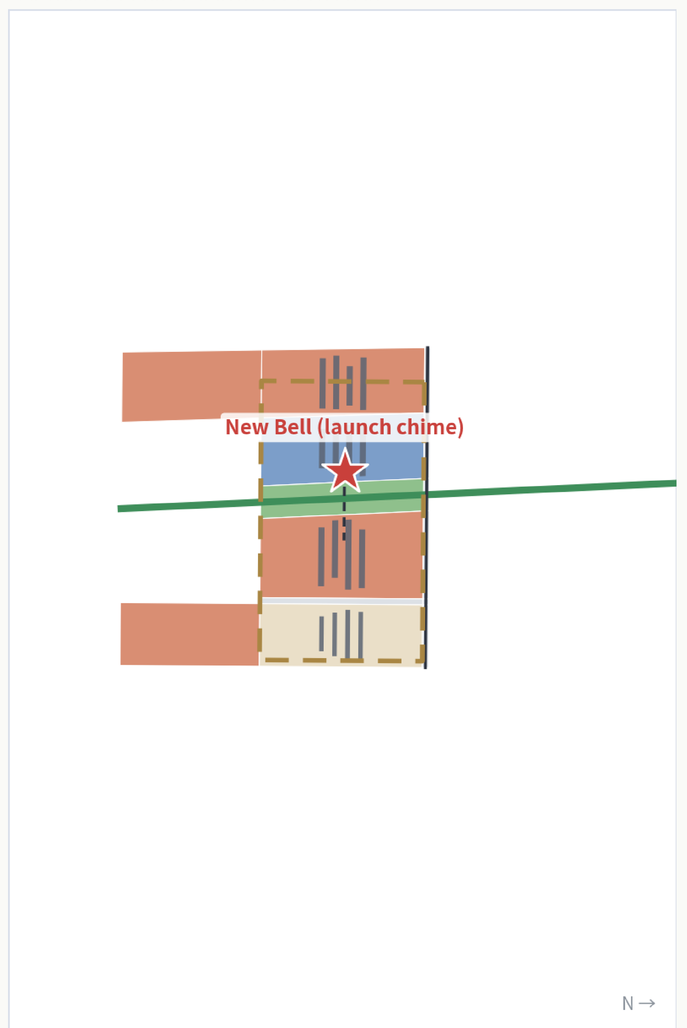
Fig.3 • Three key areas in detail: REN • CONG • ZHONG

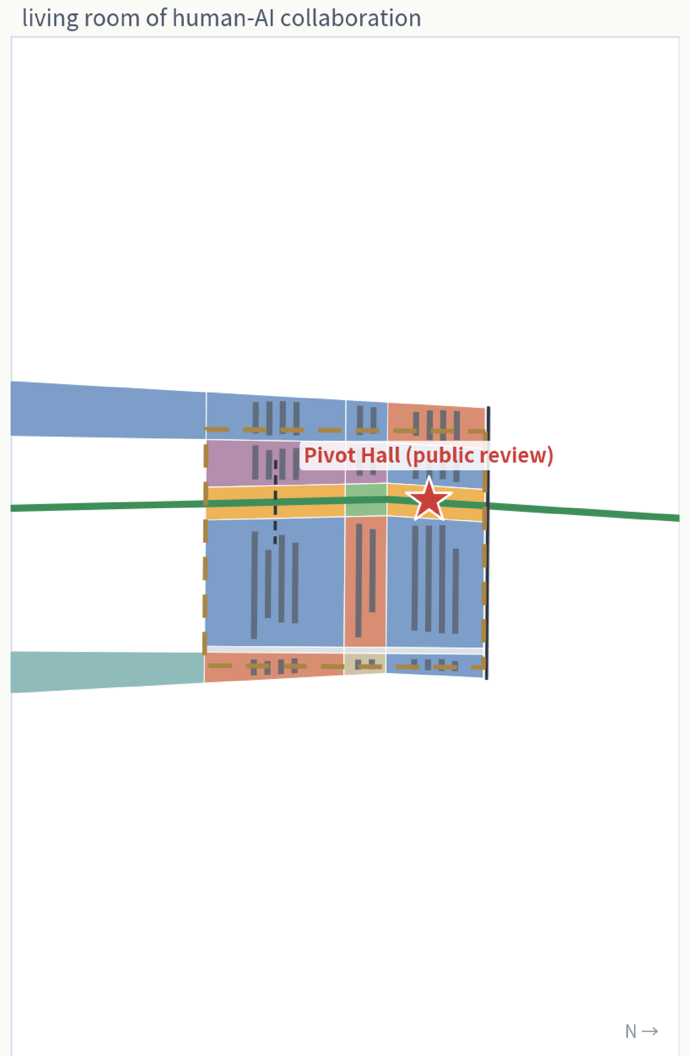
Three ways of climbing on one spine (provisional rectangles; official polygons pending)

① Dazhongsi • 人
AI-native retail & business renewal



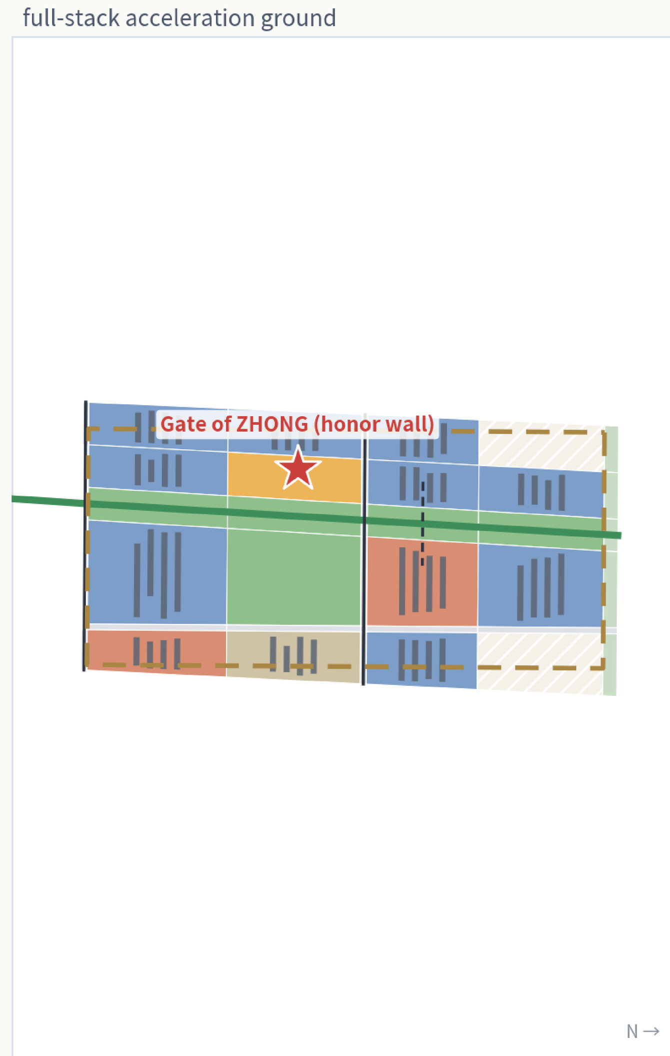
provisional 72.0 ha (announced ≈ 72.0 ha)
launch bell • station stitching • adaptive reuse

② Origin Community • 从
living room of human-AI collaboration



provisional 104.3 ha (announced ≈ 104.3 ha)
station narrative • public review • open source

③ Zhongzhiyuan • 众
full-stack acceleration ground



provisional 192.9 ha (announced ≈ 192.1 ha)
research clusters • test scenarios • Qinghe reserve

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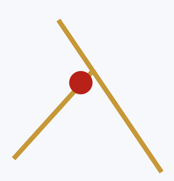
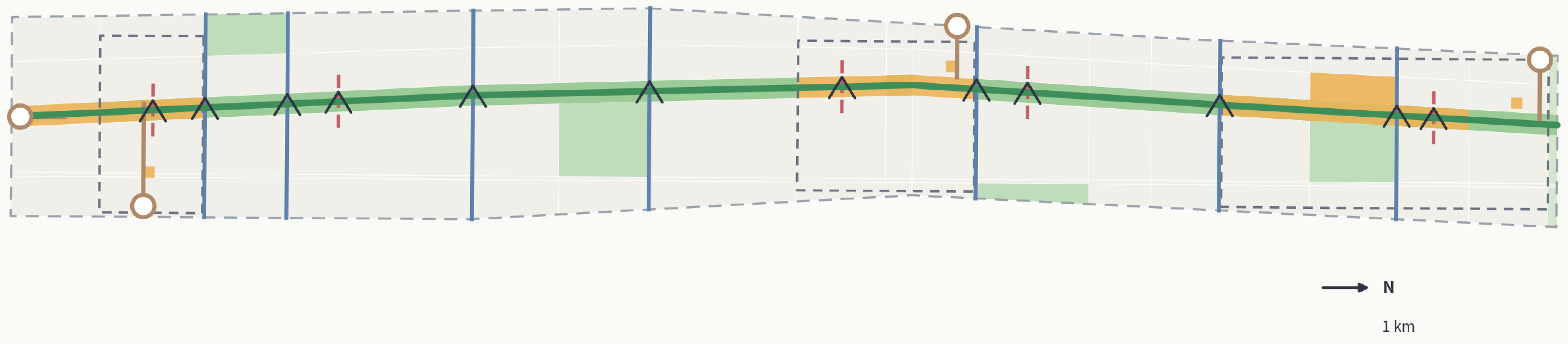


Fig.4 • Mobility and blue-green space: twelve stitches

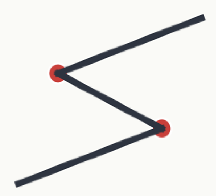
Re-stitching the two sides a century of rail once cut apart



Xiaoyuehe blue-green corridor (east wing)

- Vehicular stitch (existing street ref.)
- Pedestrian stitch (new concept)
- Slow spine (heritage park greenway)
- Transit link

12 stitches | 4 transit links | slow spine ~9.7 km | green ratio 15.4%



Standard stitch section: switchback acce
(slope ≤1:12 • two-way wheelchair pas

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Fig.5 • Metrics and evidence chain: every number recomputable, every gap declared

GeoJSON → EPSG:4548 → metrics.json → drawings/HTML → four self-check gates

Recomputed metrics (provisional basis)

Metric	Value	Note
Site area (recomputed)	1,141.3 ha	announced ≈ 1,140 ha
Green ratio (concept)	15.4%	park spine + pockets + buffer
Public space ratio	5.4%	plazas + active park + forecourts
Street-land ratio	3.8%	stitch corridors (not redlines)
Concept building volume	6.46 M m^2	low-confidence design massing
FAR (statutory)	pending	no approved controls — not invented
Height (statutory)	pending	aviation/heritage limits unpublished

Evidence chain

Provisional boundaries
↓
11 design layers geometry/*.geojson
↓
EPSG:4548 equal-area calc
↓
metrics.json 26+ metrics
↓
Drawings • HTML • PDF explanation layer
↓
Four gates self_check.json

Pending official data

- ☐ Official polygons for all three scopes and key areas
- ☐ Statutory controls (FAR / height / density / green / setback)
- ☐ Existing building stock, tenure, retain-renovate data
- ☐ Road redlines, station footprints, municipal capacity
- ☐ Heritage protection zone GIS layers

Once official data arrives: the whole package is regenerated — no single boundary file is swapped in isolation.

Coverage: announcement 1.3/1.4/1.5 + agent.1–6 = 23 items → compliance_

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Three AI Pilgrimage Landmarks (south to north)

- New Voice of the Great Bell (Go-Live Bell Platform) — every public AI service tolls its go-live and exit; citizens can hear it.
- REN Pivot Station (Public Review Hall) — purpose, data boundary, review records, and exit mechanism of every scenario, open to consultation.
- Gate of the Many (Contributor Honor Wall) — community-adopted contributions engraved in real time, including every participant of this open call.

Gradient-Graded Governance (G1/G2/G3)

- G1 instant human takeover: 8 everyday scenarios — guides, service desks, health kiosks.
- G2 periodic review + one-touch degradation: demand-responsive lighting, elder companionship.
- G3 closed test segments: low-speed shuttle, robot delivery, shared test field — the 3 industry testing scenarios.
- Every scenario card carries a REN fallback plan: when to degrade to human, how to exit, who reviews.

Three Phases (conceptual)

- Near 2026–2028, Through-Running and Launch: spine opened, 12 REN clasps, pivot plaza and Public Review Hall (~383.7 ha).
- Mid 2029–2031, Acceleration and Transformation: Zhongzhiyuan clusters, Dazhongsi consumption transformation, two test segments (~293.6 ha).
- Long 2032–2035, Networking and Synergy: housing-service band renewal, wing interfaces matured, reserves activated (~464.0 ha).